

ANIRBAN GHOSH ARGHA

Full Stack Developer | DevOps & Cloud-Native Engineering | Scalable Backend & Database Systems

Rajshahi, Bangladesh (UTC+6) · Open to Remote & Relocation · Available immediately · anirbanargha0053@gmail.com
github.com/cryptic0053 · linkedin.com/in/anirban-argha · portfolio-cryptic0053s-projects.vercel.app

PROFESSIONAL SUMMARY

Computer Science & Engineering graduate of KUET (B.Sc., 2026) who designs, ships and operates full-stack, cloud-native software. I build React / Next.js front-ends against Django REST and FastAPI services, containerise them with Docker, orchestrate them on Kubernetes, and deliver them through GitHub Actions pipelines with Prometheus and Grafana observability. Certified as a Fundamental IT Engineer under the ITEE / ITPEC framework (April 2026), and author of an A+ undergraduate thesis on scalable reverse top-k query processing. Comfortable owning a feature end to end and working asynchronously with distributed teams across time zones.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, C++, Java, C, PHP, SQL, Swift
- **Frontend:** React, Next.js, Vue.js, Tailwind CSS, Bootstrap, ShadCN UI, Zustand, Vite
- **Backend:** Django, Django REST Framework, FastAPI, Node.js / Express, Laravel, REST APIs, JWT & RBAC, Gunicorn
- **DevOps & Cloud:** Docker, Kubernetes (Minikube, rolling updates, health probes), AWS Lambda, GitHub Actions, CI/CD, Prometheus, Grafana, SonarQube, Vercel, Render
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, Streamlit, Jupyter
- **Tools & Practices:** Git, Linux, Postman, Swagger / OpenAPI, Microservices, System Design, High Availability, OOP, Data Structures & Algorithms

CERTIFICATIONS

Fundamental IT Engineer (FE) — ITEE Level 2

April 2026

Bangladesh IT-engineers Examination Center (BD-ITEC), Bangladesh Computer Council

- Awarded **Full Passer** status in the April 2026 national sitting (Examinee No. FE04-0007). The Information Technology Engineers Examination is conducted under the ITPEC framework and is aligned with Japan's IPA IT Engineers Examination, covering algorithms, databases, networks, security and system architecture.

Mastering DevOps: From Fundamentals to Advanced Practices

Ostad · 2025

Certificate of Completion — Credential ID C29273

- CI/CD pipeline design, Docker, Kubernetes and cloud infrastructure operations.

Full Stack Web Development with Python, Django & React

Ostad · 2025

Certificate of Completion — Credential ID C28956

- Building and deploying scalable applications with Django REST Framework and React.

SELECTED PROJECTS

Finch-EC — Cloud-Native E-Commerce Platform

github.com/cryptic0053/Finch-EC

Vue.js · Node.js / Express · Docker · Kubernetes · GitHub Actions · Prometheus · Grafana

- Built a decoupled storefront and API as independently containerised services, deployed with production-grade Kubernetes manifests (Deployments, Services, Ingress).
- Automated build, image push and rollout through GitHub Actions; managed credentials with Kubernetes Secrets rather than baked-in configuration.
- Instrumented the cluster with Prometheus and Grafana for real-time health and resource dashboards.

Learning Management System — Full-Stack Education Platform

github.com/cryptic0053/Learning-Management-System

React (Vite) · Zustand · Tailwind CSS · ShadCN UI · Django REST Framework · PostgreSQL · Render

- Implemented role-based access control with separate Student and Teacher dashboards and permissions, secured by JWT authentication.
- Delivered course authoring, enrolment and real-time progress tracking, with a responsive UI and Swagger-documented API.
- Deployed to Render with automated database provisioning; released under MIT licence.

Zero-Downtime Deployment on Kubernetes

github.com/cryptic0053/zero-downtime-app

Django · Gunicorn · Docker · Kubernetes · Minikube

- Engineered a highly available Django service that ships new versions with no dropped requests, using Kubernetes rolling updates plus readiness and liveness probes.
- Verified continuity empirically by polling a version endpoint throughout a live rollout and confirming uninterrupted responses.

Serverless FastAPI on AWS Lambda

github.com/cryptic0053/fastapi_lambda

FastAPI · AWS Lambda · Mangum · Docker · Bash

- Authored a production-ready template that runs FastAPI on Lambda via the Mangum ASGI adapter, scaling automatically and costing nothing while idle.
- Supports two packaging paths — container images for heavy dependencies such as pandas, and zip bundles for lightweight APIs — with scripted build tooling.

Developer Portfolio Website

portfolio-cryptic0053s-projects.vercel.app

Next.js · TypeScript · Tailwind CSS · Vercel

- Designed and shipped a typed, responsive personal site presenting research, projects and skills, deployed continuously to Vercel from GitHub.

Movie Recommender System

github.com/cryptic0053/Movie-Recommender-System

Python · Scikit-learn · Pandas · NumPy · NLTK · Streamlit · TMDB API

- Built a content-based recommender over the 5,000-title TMDB dataset, engineering tag vectors from genres, keywords, cast and crew with CountVectorizer and ranking neighbours by cosine similarity.
- Wrapped the model in a Streamlit interface that fetches live posters and metadata from the TMDB API.

RESEARCH

HYDART-RQ: A Hybrid Framework for Durable Reverse Top-k Query Processing over Time-Varying Preferences

B.Sc. Thesis · Grade A+

Supervisor: Dr. K. M. Azharul Hasan, Professor, Dept. of CSE, KUET

- Designed a hybrid framework for **durable** reverse top-k queries — results that remain valid as user preference vectors drift over time — rather than recomputing answers from scratch at every timestamp.
- Targeted the scalability bottleneck of classical reverse top-k processing on large, time-varying preference datasets. Awarded **A+** across both thesis terms (4.50 credits).

EDUCATION

Khulna University of Engineering & Technology (KUET)

2022 – 2026

B.Sc. in Computer Science and Engineering — Khulna, Bangladesh | CGPA 3.23 / 4.00 | 161.25 / 161.25 credits earned

- Degree requirements completed in full; all coursework, thesis and credit-group requirements satisfied. Graded **A+** in Project & Thesis and in Technical Writing & Seminar.
- Coursework: Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems, Machine Learning, AI, Image Processing & Computer Vision, Computer Graphics, Software Engineering.

Rajshahi College — Higher Secondary Certificate (HSC)

2018 – 2020

GPA 5.00 / 5.00

Rajshahi Collegiate School — Secondary School Certificate (SSC)

2018

GPA 5.00 / 5.00

AWARDS & LEADERSHIP

1st Runner-Up — 3-Minute Thesis (3MT) Segment

2025

SciBlitz 1.0, organised by the IEEE CUET Student Branch (IEEE Bangladesh Section)

- Placed second overall in a national research-communication contest, presenting scholarly work concisely and effectively within a strict three-minute format, as a team of three.

Executive Member — BITFEST 2025

3–4 January 2025

Organizing Committee, Department of CSE, KUET

- Recognised by the Head of Department and the Organizing Committee for leadership and dedication in delivering BITFEST 2025, a national inter-university technology festival.

ADDITIONAL

- Languages:** English (professional working proficiency), Bengali (native).
- Remote readiness:** Experienced with fully asynchronous Git-based workflows, pull-request review and written documentation; available for overlap with European and Asia-Pacific business hours.

REFERENCES

Dr. K. M. Azharul Hasan

Professor, Dept. of Computer Science and Engineering
Khulna University of Engineering & Technology
Undergraduate thesis supervisor
az@cse.kuet.ac.bd

Md Mehrab Hossain Opi

Lecturer, Dept. of Computer Science and Engineering
Khulna University of Engineering & Technology
opi@cse.kuet.ac.bd